

Cisco Port Security Experiment Guide

<https://gemini.google.com/app/297bd008378ef4e2>

User prompt: What is this expt about describe step by step so I can solve it nd see all outcomes

Response: This experiment demonstrates **Switch Port Security in Cisco Packet Tracer** to restrict access based on MAC addresses and analyze how a switch reacts under three distinct violation modes: **Protect**, **Restrict**, and **Shutdown**.

Step-by-Step Execution Guide

Step 1: Network Setup

1. Add a **2960 Switch** and connect several authorized **PCs** (e.g., PC0 to Fa0/1, PC1 to Fa0/2, PC2 to Fa0/3, PC3 to Fa0/4).
2. Add an unauthorized device (labeled **Rogue PC** or PC4).
3. Assign static IP addresses (e.g., 192.168.1.X /24) to all end devices.

Step 2: Configure Port Security Modes on the Switch

Open the Switch **CLI** (enable -> configure terminal) and configure each port:

- **Port 1 (Fa0/1) - Shutdown Mode (Default)**

```
interface FastEthernet0/1
  switchport mode access
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
exit
```

- **Port 2 (Fa0/2) - Protect Mode**

```
interface FastEthernet0/2
  switchport mode access
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation protect
exit
```

- **Port 3 (Fa0/3) - Restrict Mode**

```
interface FastEthernet0/3
  switchport mode access
  switchport port-security
  switchport port-security maximum 1
```

```

switchport port-security mac-address sticky
switchport port-security violation restrict
exit

```

Step 3: Register Authorized MAC Addresses

1. Ping from each authorized PC to any other PC in the network.
2. The switch automatically learns and saves each authorized MAC address dynamically via the `sticky` command.

Step 4: Trigger the Security Violations

1. Disconnect the link cable from an authorized port (e.g., Fa0/1, Fa0/2, or Fa0/3).
2. Reconnect that cable to the **Rogue PC**.
3. Attempt to ping any device from the **Rogue PC**.

Expected Outcomes Matrix

Violation Mode	Traffic Dropped?	Violation Counter Incremented?	Port Status	Recovery Action
Protect	Yes	No (Stays at 0)	Stays Up (Green)	Reconnect authorized device; traffic resumes immediately.
Restrict	Yes	Yes (Increments on show port-security)	Stays Up (Green)	Generates syslog alert. Reconnect authorized device; traffic resumes.
Shutdown	Yes	Yes	Goes Down (err-disabled / Red)	Requires manually running shutdown then no shutdown on the interface.

Verification Commands

Run these in Privileged EXEC mode (Switch#):

- `show port-security` — View summary status and violation counts across ports.
- `show port-security interface fastEthernet 0/1` — View detailed MAC limits, sticky entries, and current security state for a specific interface.
- `show mac-address-table` — Verify sticky MAC address entries bound to switch ports.

Need help resolving an `err-disabled` state or testing a specific ping path?

User prompt: Should I assign IP first or get the switch cli config done first

Response: Assign the IP addresses to the PCs **first**, before configuring port security on the switch.

Here is why and how the sequence works:

1. **Assign IP Addresses First:** The switch learns authorized MAC addresses dynamically when traffic flows through a port using the `switchport port-security mac-address sticky` command. For a PC to send packets (like an ICMP ping) so the switch can "learn" its MAC address, the PC must already have a valid IP address configured.
2. **Configure Port Security Second:** Enter the CLI commands on the switch to enable port security, set maximum MAC limits, and define violation modes (`shutdown`, `protect`, `restrict`).
3. **Generate Traffic Third:** Send a ping from each PC. The switch will capture the source MAC address, bind it as a `sticky` address, and actively enforce security from that point forward.

If you configure port security first without assigning IPs, nothing breaks, but the switch won't actually learn or lock in the allowed MAC addresses until the PCs have IPs and send their first packet.

User prompt: tell me the outcome or change to notice and what to do to notice for the 3 violation modes

Response:

1 Test Protect Mode

Port configured with 'protect'

Connect the Rogue PC to Fa0/2 and send a ping command (`ping 192.168.1.1`).

- **Outcome / Change to Notice:** The ping will fail (time out). The link light remains **green** (up), and running `show port-security interface fa0/2` shows the Security Violation count stays strictly at **0**.
- **How to verify success:** Reconnect the original PC and ping again; traffic resumes instantly without touching the switch CLI.

2 Test Restrict Mode

Port configured with 'restrict'

Connect the Rogue PC to Fa0/3 and attempt to ping.

- **Outcome / Change to Notice:** The ping will fail, but the switch generates a syslog console alert notifying you of a security violation. The link light stays **green**, but the violation counter actively increments.

- **How to verify success:** Run `show port-security interface fa0/3` in the switch CLI. You will see the **SecurityViolation Count** increase (e.g., from 0 to 1 or more).

3 Test Shutdown Mode

Port configured with 'shutdown'

Connect the Rogue PC to Fa0/1 and attempt to ping.

- **Outcome / Change to Notice:** The port instantly disables itself. The link light on the switch turns **red**, and a syslog message reports the interface state changed to down (err-disabled).
- **How to verify success:** Run `show interface fa0/1` in the switch CLI. The output will explicitly display `FastEthernet0/1 is down, line protocol is down (err-disabled)`.

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User prompt: So for the shutdown one if I connect the rogue pc the connection shuts down now when I connect back the original connected pc with FastEth0/1 will it still stay off ? as it is shutdown mode, conclude this scenario for all cases

Response: **Yes, for the Shutdown mode, it will stay off.** Even when you unplug the Rogue PC and plug the original, authorized PC back into Fa0/1, the interface remains completely **disabled (Red)** until an administrator manually resets the port via the switch CLI.

Here is the exact conclusion for what happens across all three violation modes when you reconnect the original authorized PC:

- **Shutdown Mode (Fa0/1):** The port stays **Down (Red)**. Plugs, pings, and traffic will fail for the original PC until you manually execute `shutdown` followed by `no shutdown` on the interface.
 - **Restrict Mode (Fa0/3):** The port turns **Active (Green)** immediately. The original PC regains network connectivity right away, but the violation history counter remains saved in the switch memory until cleared.
 - **Protect Mode (Fa0/2):** The port turns **Active (Green)** immediately. Traffic drops for the Rogue PC and instantly resumes for the original PC without leaving any violation log entries.
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User prompt: By turns green do u mean from red to green or it stays green all the time even when connections are removed from the original pcs and made with the rogue pcs? for: Restrict Mode (Fa0/3): The port turns Active (Green) immediately. The original PC regains network connectivity right away, but the violation history counter remains saved in the switch memory until cleared. Protect Mode (Fa0/2): The port turns Active (Green) immediately. Traffic drops for the Rogue PC and instantly resumes for the original PC without leaving any violation log entries.

Response: It transitions back from Amber/Red to Green.

Here is exactly what happens to the physical link lights during each phase of unplugging and swapping cables in Packet Tracer:

- **When you unplug the original PC:** The link light goes **completely off** (no color/gray) because physical layer connectivity is lost when the cable is removed.
- **When you plug in the Rogue PC:** The link light initially turns **Amber** for ~30 seconds (spanning tree protocol calculation) before turning **Green**. In both **Protect** and **Restrict** modes, the link light **stays Green** while connected to the Rogue PC—it never turns red. Physical link is up, but the switch silently blocks/drops the rogue traffic at the data layer.
- **When you reconnect the original PC:** The light goes through the standard link-up sequence (Amber to Green). Once **Green**, authorized traffic flows immediately.

The only mode where the light actually turns **Red** (and stays Red even after plugging the original PC back in) is **Shutdown** mode.

User prompt: Which command to give and check for after effects from cli?

Response: To test and verify the after-effects for all three violation modes from the switch CLI, execute these specific diagnostic and verification commands:

1 1. Check Global & Port Security Status

All Modes

Run this command in Privileged EXEC mode (Switch#) to view a summary of all secure ports, current MAC counts, violation counters, and active security actions:

```
show port-security
```

- **How to verify:** Compare the SecurityAction column (Shutdown, Protect, or Restrict) and check the SecurityViolation column to see which ports have recorded violations.

2 2. Inspect Specific Port Details

Protect & Restrict Modes

Run this command for a specific interface to inspect its detailed status:

```
show port-security interface fastEthernet 0/3
```

- **How to verify:** Look at **Port Status** (should show Secure-up) and **SecurityViolation Count**. In **Restrict** mode, this count increments with every blocked ping packet. In **Protect** mode, it remains 0.

3 3. Verify Err-Disabled State

Shutdown Mode

Check the physical and operational status of a tripped interface (e.g., Fa0/1):

```
show interface fastEthernet 0/1
```

- **How to verify:** The first line of output will display: FastEthernet0/1 is down, line protocol is down (err-disabled)

4 4. Reset a Shutdown Port

Recovery

To restore a port disabled by **Shutdown** mode, reconnect the valid PC and execute:

```
configure terminal
interface fastEthernet 0/1
shutdown
no shutdown
exit
```

- **How to verify:** Run `show interface fastEthernet 0/1`. The output should change to FastEthernet0/1 is up, line protocol is up, and the link light in Packet Tracer will turn green.

User prompt: Switch>enable Switch#configure terminal Enter configuration commands, one per line. End with CNTL/Z. Switch (config) #show port-security % Invalid input detected at '^' marker. Switch (config) # >

Response: The show commands cannot be run directly from Global Configuration mode (Switch(config)#) unless you use the do prefix.

To fix this error, you have two simple options:

Option 1: Use the do keyword (Recommended)

You can run any EXEC-level show command inside configuration mode by placing **do** in front of it:

```
Switch(config)# do show port-security
```

Option 2: Exit to Privileged EXEC Mode

Exit back to the Switch# prompt first, then run the command directly:

```
Switch(config)# exit
Switch# show port-security
```

User prompt: how to ping from rogue pc?

Response: To ping from the Rogue PC in Cisco Packet Tracer:

1 1. Open Command Prompt on Rogue PC

Open Desktop Tab

Click on the **Rogue PC** icon in your topology workspace. In the popup window, select the **Desktop** tab and click on **Command Prompt**.

2 2. Send Ping Command

Execute ICMP Packet

Type `ping` followed by the IP address of any authorized PC connected to the switch (e.g., 192.168.1.1):

```
ping 192.168.1.1
```

Press **Enter**.

3 3. Verify the Traffic Dropping

Observe Output

- **Expected Result:** The output will display **Request timed out** for all 4 packets because port security blocks the unauthorized MAC address.
 - **Verify Success:** Check the switch CLI (`do show port-security`). Depending on the mode:
 - **Protect:** Packet dropped quietly.
 - **Restrict:** Packet dropped + violation counter increments.
 - **Shutdown:** Port immediately turns off (Red light).
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